

KINLOSS, United Kingdom

WMO: 030660

Lat: 57.6456N

Lon: 3.5636W

Elev: 22

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.1	26.7	19.5	14.7	26.2	22.4	16.9	30.2	36.4	44.0	32.9	42.6	3.6	230	0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	72.9	62.0	69.6	60.4	67.1	58.9	63.7	70.3	61.8	67.7	60.1	65.5	12.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.8	79.7	67.1	59.1	75.0	64.5	57.5	70.6	62.9	29.0	70.2	27.6	67.7	26.5	65.7	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.9	25.3	22.4	DB	17.6	80.1	4.9	4.0	14.1	83.0	11.2	85.4	8.4	87.6	4.8	90.6	(4)
(5)			WB	16.9	67.3	4.9	1.8	13.4	68.6	10.5	69.7	7.7	70.7	4.1	72.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	39.3	39.8	42.3	46.1	50.2	54.9	58.8	58.3	54.7	49.3	43.6	39.4
	DBStd	8.72	5.46	5.59	5.31	5.18	5.07	4.40	4.29	4.00	4.28	5.08	5.53	6.81
	HDD50	1687	333	288	245	139	60	7	0	0	8	74	200	333
	HDD65	6178	797	704	705	568	460	302	198	213	309	485	642	795
	CDD50	997	1	3	5	22	65	155	274	257	150	53	9	3
	CDD65	14	0	0	0	0	0	1	8	4	1	0	0	0
	CDH74	79	0	0	0	0	2	6	43	27	1	0	0	0
	CDH80	7	0	0	0	0	0	0	4	3	0	0	0	0
Wind	WSAvg	10.4	11.4	11.8	11.8	10.8	9.8	9.6	8.7	8.9	9.9	10.4	10.9	10.7
Precipitation	PrecAvg	24.8	2.1	1.6	1.7	1.5	1.8	2.1	2.2	2.6	2.4	2.4	2.4	2.0
	PrecMax	33.5	4.4	4.2	3.1	5.1	4.5	6.3	5.9	6.6	11.8	5.1	4.3	4.1
	PrecMin	13.6	0.3	0.2	0.5	0.2	0.2	0.6	0.4	0.7	0.7	0.4	0.6	0.5
	PrecStd	4.0	1.1	0.9	0.8	0.9	0.9	1.2	1.1	1.4	1.7	1.2	1.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	56.9	59.3	65.6	71.2	73.8	79.4	77.6	71.9	64.7	59.3	57.1
		MCWB	49.8	51.0	50.5	54.0	58.3	61.4	64.9	64.1	61.7	57.3	53.8	51.9
	2%	DB	52.2	54.3	55.4	61.0	65.9	68.6	73.5	71.6	67.7	62.1	56.4	54.6
		MCWB	47.5	49.2	48.8	51.9	56.1	58.7	62.7	61.9	59.6	55.6	51.4	50.2
	5%	DB	50.1	51.7	53.0	57.9	62.4	66.0	69.8	68.3	64.7	59.8	54.3	52.0
		MCWB	45.9	46.8	47.2	50.1	54.1	57.5	61.1	60.0	57.5	54.0	49.7	47.9
	10%	DB	47.6	48.8	50.9	55.1	59.1	63.1	66.6	65.5	62.3	57.5	52.1	49.5
		MCWB	43.9	44.3	45.8	48.5	52.1	55.8	59.3	58.4	55.8	52.3	48.2	45.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	52.1	51.9	55.1	61.0	63.2	66.8	65.9	62.7	58.4	54.6	52.4
		MCDB	54.3	56.2	57.5	63.1	68.6	71.0	75.1	74.4	69.6	63.2	58.6	56.4
	2%	WB	48.1	49.3	49.5	52.8	57.1	60.1	64.2	63.1	60.3	56.3	52.1	50.3
		MCDB	51.7	53.7	54.5	59.8	63.5	66.6	70.7	69.3	66.3	61.0	55.7	54.1
	5%	WB	46.2	47.2	47.7	50.7	54.8	58.2	62.2	61.2	58.4	54.7	50.1	48.2
		MCDB	49.7	51.4	52.5	56.8	60.6	64.2	68.0	66.7	63.3	58.9	53.5	51.6
	10%	WB	44.1	44.6	46.0	49.0	52.8	56.6	60.4	59.5	56.8	53.0	48.4	45.8
		MCDB	47.5	48.3	50.6	54.4	58.3	62.1	65.5	64.1	61.1	56.7	51.7	49.2
Mean Daily Temperature Range	5% DB	MDBR	9.1	10.7	11.8	13.3	14.1	12.8	12.9	13.0	12.4	11.5	10.1	9.9
		MCDBR	11.0	12.2	14.7	17.7	19.9	17.7	18.5	17.4	16.2	13.9	12.3	11.9
		MCWBR	8.9	9.2	10.1	10.7	11.7	10.1	10.1	9.9	10.1	9.8	9.9	10.0
	5% WB	MCDBR	10.8	11.6	13.8	16.6	18.3	16.2	16.6	16.0	14.9	13.2	12.0	12.0
		MCWBR	9.1	9.4	10.0	10.6	11.6	9.9	10.1	9.9	10.2	10.0	10.1	10.4
Clear-Sky Solar Irradiance	taub	0.307	0.331	0.355	0.374	0.375	0.372	0.375	0.363	0.349	0.334	0.309	0.293	
	taud	2.246	2.340	2.337	2.321	2.364	2.400	2.400	2.441	2.483	2.443	2.298	2.177	
	Ebn at Noon	172	225	251	265	273	275	271	267	253	222	170	137	
	Edh at Noon	16	23	29	34	35	34	34	30	25	20	15	12	
All-Sky Solar Radiation	RadAvg	134	335	665	1093	1474	1447	1400	1104	771	407	174	91	
	RadStd	17	33	71	103	80	121	116	70	69	35	14	10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page